

Total Hand Orientation Reinforcement Glove for Assisted Mobility and Strength

Charlie Scheel, *Member, IEEE*, Mark Dannemiller, *Member, IEEE*, and Dashiell Yount

Abstract— The Total Hand Orientation Reinforcement (T.H.O.R.) Glove is an assistive device targeting the user's finger mobility and grip strength. It utilizes force sensing resistors at the fingertips to determine the user's intended movement. The processed force measurement is used to determine the desired change in velocity of the glove digits, which is powered via pairs of servo motors for flexion and extension. These motors control the tension of a pulley system, supporting the user's motion with the glove, even against an opposing force, such as a heavy object. This novel assisted mobility device has applications in hand-oriented jobs where a worker's grip strength is stressed over an extended period.



Figure 1: THOR Concept

Index Terms—Assistive Force, Mobility Device

I. INTRODUCTION

THE Total Hand Orientation Reinforcement (T.H.O.R.)

Glove is an assistive device designed to improve finger mobility and grip strength. By using force sensing resistors (FSRs), the glove detects pressure applied by the user's fingers and translates this force into control signals for servo motors. [2] These motors drive a pulley system to assist in flexing and extending the fingers, helping the user grip objects with enhanced strength and support, even when facing resistance. [3] This device is particularly useful in tasks that demand prolonged hand strength and dexterity. This paper will describe the processes, design work, testing, and implementation involved in developing the T.H.O.R. Glove, focusing on the integration of mechanical, electronic, control systems, and how each component contributes to its functionality and overall design.

II. MECHANICAL DESIGN

The THOR glove is composed of a comfortable inner layer which contacts the user's skin, a flexible rubber layer which defines the glove's structure, and the tendon system comprising of motorized actuators, tendons, and tendon sheaths guided to the finger tips of the device. These aspects enable 16 pounds of additive grip strength to the user and has been designed for prolonged use for a standard work day.

A. Anatomical and Mechanical Challenges

The primary goal of the THOR glove is to assist in the flexion and extension of the user's fingers. This movement is just one of several motions that the human hand is capable of performing, and therefore some constraints must be applied to the user's hand in order to ensure successful assistance by the actuator, as shown in Figure 2 below.

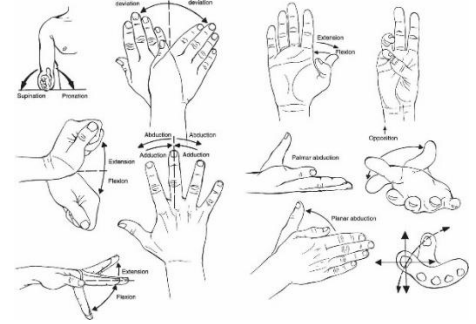


Figure 2: Human Hand Movements [1]

The first constraint is on rotation of the wrist. In the human hand, the fingers are adjusted based on changes to the wrist rotation in order to maintain position. The glove would have to do the same via sensing these wrist rotations, which requires additional complexity. To solve this challenge, the THOR glove would need to hold the user's wrist static, preventing flexion/extension and radial/ulnar deviation. Our first prototype does not accomplish this feature.

The second constraint is on the final knuckle of each finger. As displayed in Figure 7, the operation of the finger motion system is reliant on the final knuckle having freedom of movement from the assisted system. This freedom allows the user to signal intention from within the glove and command it to flex/extend. There is no freedom in the closing motion of the finger, but the opening motion engages a TPU spring and

presses on an FSR sensor. [10] This allows the device to respond to the user opening hand motion.

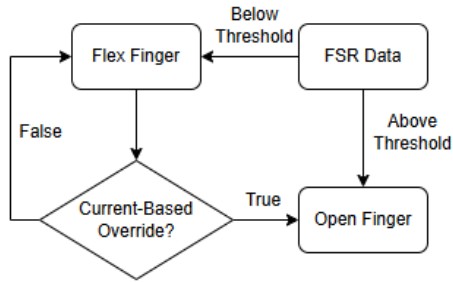


Figure 3: Sensor Configuration in Mechanical Design

Besides constraints to the human anatomy, there are several other mechanical requirements deemed necessary for this project:

- Mechanical stop to prevent overextension.
- Securing joints of glove tightly
- Holding adequate weight beyond user's normal grip strength.

B. Structure of THOR Glove

The THOR Glove's mechanical architecture was conceived with three priorities in mind: wearer comfort, reliable force transmission, and straightforward serviceability. At its core, the glove pairs a soft, breathable knit liner with a flexible thermoplastic-polyurethane (TPU) exoshell. The liner hugs the hand, manages perspiration, and prevents skin abrasion, while the exoshell provides the rigid anchor points needed to guide the tendon sheaths without restricting natural joint motion. Each finger and the thumb carries an external 100 lb fishing line tendon that slips through low-friction TPU routes. By routing the tendons outside the hand, the design avoids the bulk and heat-buildup often associated with internal cable paths and leaves the palmar surface free for tactile feedback.

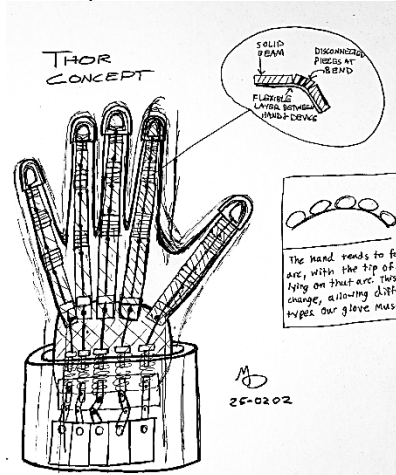


Figure 3: Wireframe of Glove Components

All five tendons converge at the wrist-mounted actuator pod, a rounded housing that wraps partially around the distal forearm. Inside this pod, five high-torque miniature servomotors—See Figure 4—sit in a radial pattern on a custom four-layer control PCB. The board not only secures the motors mechanically but also shortens power and signal paths, minimizing electrical noise and reducing harness complexity. Tendon terminations are oriented so that a 15-degree commanded angle corresponds to a completely relaxed cable, whereas angles of 200 degrees or more pull the tendon to its fully contracted state. For fingers whose servos rotate in the opposite direction, a simple software inversion ensures that the wearer experiences consistent motion regardless of motor orientation.

Power delivery and additional regulation are handled outside the pod in a dedicated battery pack that contains its own two-layer power-distribution PCB. A twelve-conductor harness—with 22-AWG silicone-insulated wires bundled in an expandable PET braid—bridges the pack to the pod. This cable carries isolated 6.8 V and 5 V rails and multiple ground returns while maintaining enough flexibility to follow wrist articulation without transmitting torque to the glove. Keyed circular connectors and heat-shrink strain relief at both ends let users decouple the electronics from the soft goods in seconds, streamlining maintenance and upgrades.

Safety has been engineered into the mechanics from the outset. Each tendon path incorporates a physical over-extension stop molded into the TPU guides, preventing undue strain on the servos or the wearer's joints. The actuator pod distributes its 150-gram mass over a broad surface area and secures with a hook-and-loop cuff, keeping the center of gravity close to the wrist to avoid user fatigue. Together, these elements produce a lightweight assembly—less than 400 grams on the hand, excluding the off-body battery—that can withstand more than a thousand full-stroke cycles without mechanical failure while delivering in excess of 10 newtons of additive grip force per finger.

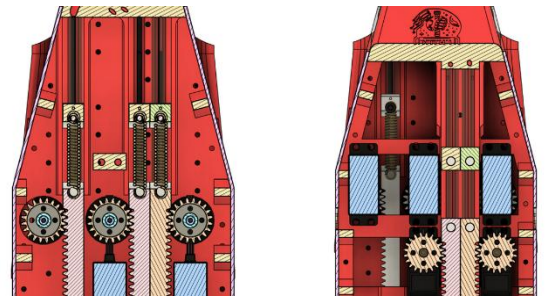


Figure 4: Actuator Design

III. T.H.O.R. CONTROL SYSTEM

The THOR control system consists of one microcontroller operating the

A. Teensy 4.0

The Teensy 4.0 microcontroller was selected for this project due to its exceptional performance characteristics, including low power consumption, a robust array of analog-to-digital converter (ADC) ports, and its compact form factor. Specifically, the Teensy 4.0 is powered by a 600 MHz ARM Cortex-M7 processor, offering substantial computational capability, and includes 14 ADC pins with 16-bit resolution for high-precision analog signal acquisition. Its small footprint of 35.6 mm x 17.8 mm ensures seamless integration into space-constrained designs. These attributes make the Teensy 4.0 an optimal choice for detecting subtle finger movements through low-voltage sensing using Force Sensitive Resistors (FSRs), as its high-resolution ADCs can accurately capture minute changes in resistance. Additionally, the microcontroller's compact size and low power requirements enable it to be comfortably and efficiently mounted on the wrist, supporting wearable applications with reliable performance and minimal energy consumption.

B. Sensors

The sensor component of this design consists solely of Force Sensing Resistors (FSRs), which detect the force applied by the user's fingers. An FSR is a flexible, variable resistor that increases in resistance as pressure increases. [11] The THOR project utilizes ten FSRs, two for each finger of the glove, allowing the user to maintain natural finger movement due to their flexibility. The circuit for the FSR is illustrated in figure 2. The Teensy 4.0 microcontroller is equipped with an onboard 3.3 V regulator which is used to supply power to the FSRs. Each FSR is configured in a voltage divider circuit, with its output connected to a 5.1 k Ω resistor to ground and a wire leading to the ADC input of the Teensy 4.0 at the same node. When full force is applied, the FSR outputs a maximum of 3 V to the ADC. [2]



Figure 5: Sparkfun Force Sensing Resistor [2]

The concept operates as follows: when the user applies pressure to the fingertip sensor, the voltage at the ADC on the Teensy increases, triggering a Pulse Width Modulation

(PWM) signal to the servo motors. This causes the servos to rotate in a positive direction, enabling the user to clasp or bend their fingers. Alternatively, when the user lifts their fingers off the fingertip sensor and applies pressure to the FSR located on the top of the nail, the servo receives a signal to rotate in the opposite direction, allowing the fingers to retract.

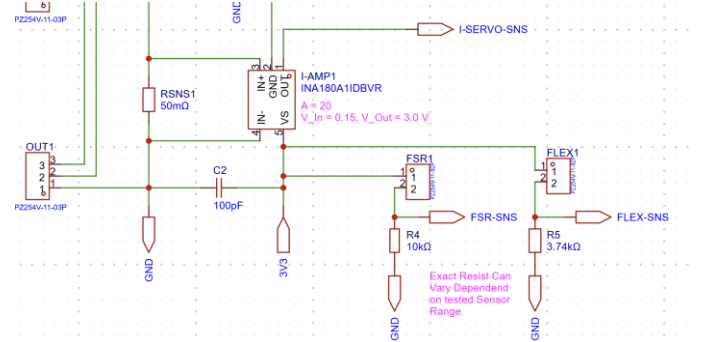


Figure 6: Sensor Circuit Schematic

C. Software

The THOR Glove firmware is implemented on a Teensy 4.0 microcontroller running at 100 Hz and organized into two main components: a high-level control sketch (lab4.ino) and a hardware abstraction layer (Finger class). This architecture cleanly separates control logic from low-level operations, ensuring modularity and ease of maintenance.

The high-level code handles initialization, calibration, mode switching, and a closed-loop control loop that adjusts each finger's position based on sensor inputs. It also enforces safety via current monitoring and includes a simple serial debug interface.

The Finger class abstracts servo control, FSR voltage sensing, current feedback, and angle limits. It maps target angles to servo pulses, optionally inverts them for symmetric behavior, and includes methods for fault release and basic sensor reads. Sensor readings are scaled to physical units; for example, FSR voltage is computed from a 12-bit ADC read, and current is derived from a sensed voltage through a known resistance.

Each control cycle reads finger inputs, computes a desired angular velocity from FSR displacement, and integrates this into a new target angle. If current exceeds safe thresholds, the finger is released to prevent damage. Global faults are triggered if total current exceeds 12 A. A debug function prints real-time values in a formatted ASCII table using ANSI escape codes.

This layered approach supports closed-loop control on a constrained embedded platform while enabling extensibility and real-time diagnostics.

D. Electronic Control

As discussed in Section III, Subsection C, the Teensy 4.0 converts the user's applied force using the Force equation: $F = A * R^{-B}$ where A and B are calibration constants. Based on the output from the Force Sensing Resistor (FSR), the Teensy 4.0 generates a Pulse Width Modulation (PWM) signal to adjust the servo motor angle, either engaging or retracting the user's fingers.

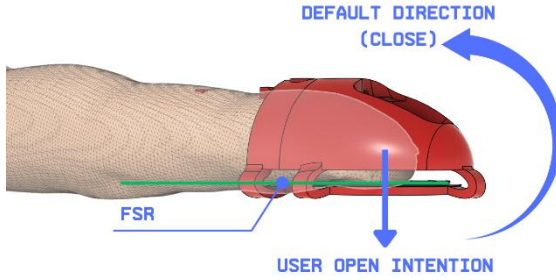


Figure 7: FSR Interaction

The servo motors operate within a voltage range of 4.8V to 6.8V, while the GPIO pins on the Teensy 4.0 output only 3.3V. To account for this, the PWM signal is sent to an R2501 optocoupler, with the opposite side connected to a 6.8V supply. A Schottky diode is also connected in series protects against back electromotive force (BEMF) that could potentially disrupt its operation.

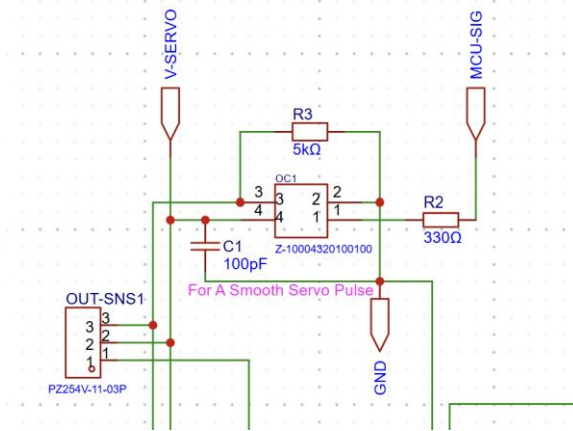


Figure 8: 25 kg Servo Control Schematic

IV. POWER PACK

The powerpack model was engineered using Fusion360 and fabricated via 3D printing with PETG-CF filament, selected for its superior strength and durability. Its

design draws inspiration from the Star Wars universe, specifically the protosaber power pack concept.

The powerpack houses a power distribution printed circuit board (PCB) designed to manage current allocation for six servo motors, each capable of drawing up to 2.2 A under stall conditions. To prevent overloading a single integrated circuit with the total 15 A current, the PCB was configured to distribute the load in 6 paths for each 25kg servo motor and to power the Teensy 4.0. Initially, the PCB incorporated LT1085CT adjustable regulators configured to deliver 6.3 V. Due to suboptimal performance, these were replaced with LM2596 Buck Converters, also set to output 6.3 V, necessitating adjustments to the PCB layout to accommodate the new components. The PCB is powered by a 11.1 V lithium-ion battery connected through an XT60 connector. But this board is capable of up to 20 V input.



Figure 9: Power Distribution PCB

The PCB output is connected by a 12-port wire cable constructed with 22-gauge wire, capable of handling up to 3 A per wire to accommodate potential servo motor stalls contained nicely in a braider cable sleeve. The cable includes five 6.3 V connections for the servo motors, one 5 V connection generated by a linear regulator to power the Teensy 4.0 microcontroller, and six ground wires to ensure balanced current distribution. This cable directly connects to the PCB which mounts the Teensy 4.0 and the opto-couplers that separate the power between the Teensy and the servo motors.

VI. FINAL PROTOTYPE

The fully integrated THOR Glove prototype (Figure 10) unites a soft-goods fabric liner, a heat-laminated TPU exoshell, an externally routed Spectra-tendon network, a PLA actuator pod, and a carbon-fiber-filled PET-G power pack into one cohesive wearable. [10] Five 25 kg·cm servomotors nest radially around the custom four-layer control PCB inside the PLA pod, keeping the assembly's mass close to the wrist and simplifying cable management. Each 0.6 mm Spectra tendon is guided exclusively through molded TPU pathways along the fingers and terminates on tied ends secured inside the pod; this arrangement lets an operator replace or re-tension a single

cable without dismantling the glove. A twelve-conductor, silicone-jacketed harness with keyed circular connectors bridges the pod to the carbon-fiber PET-G power pack, so final assembly can be completed in less than five minutes and field servicing remains straightforward.

Although the prototype meets its functional targets, several limitations point the way toward the next design iteration. The wrist cuff, printed in semi-rigid PLA, fits a broad range of users but lacks the snug, conformal grip needed for strenuous tasks; a re-engineered clamshell or Boa-dial closure will address this looseness. Under high-load grasps the exposed tendons can tug the pod forward, so the team is experimenting with Bowden-style tubes to decouple cable tension from pod placement. [4] Miniaturising the actuator pod itself—by adopting shorter-body servos or a staggered, two-tier PCB stack—will further reduce mass and improve wearer comfort. Finally, preliminary bench tests with shape-memory-alloy bundles suggest that replacing rotary servos with artificial-muscle actuators could slash weight, remove gearbox noise, and deliver a more natural compliance; this alternative actuation strategy is slated for exploration in the forthcoming revision. [8]



Figure 10: Final Product

VII. CONCLUSION

The Total Hand Orientation Reinforcement (T.H.O.R.) Glove is an innovative assistive device designed to boost finger mobility and grip strength in hand-intensive tasks. By employing force sensing resistors, servo motors, and a pulley system, it effectively translates user intent into assisted motion, supporting tasks against external resistance. The glove's ergonomic mechanical design, featuring a comfortable inner layer and flexible structure, ensures extended usability. A Teensy 4.0 microcontroller drives a precise, modular control system with real-time sensor feedback, enhancing safety and performance. The 3D-printed power pack with

optimized power distribution supports reliable operation. Testing confirms its ability to reduce fatigue and increase productivity in demanding work settings. Future enhancements, such as wrist stabilization and expanded motion capabilities, could further broaden its applications, making the T.H.O.R. Glove a valuable tool for improving worker efficiency and well-being.

REFERENCES

- [1] “Acute Wrist and Hand Injuries,” *Musculoskeletal Key*, Jun. 12 2016. [Online]. Available: <https://musculoskeletalkey.com/acute-wrist-and-hand-injuries/>. [Accessed: May 13, 2025]. ([Musculoskeletal Key](#))
- [2] SparkFun Electronics, “Force Sensitive Resistor 0.5 in (SKU SEN-09375) — product page,” Niwot, CO, USA. [Online]. Available: <https://www.sparkfun.com/force-sensitive-resistor-0-5.html>. [Accessed: May 13, 2025]. ([SparkFun](#))
- [3] P. Polygerinos *et al.*, “Towards a soft pneumatic glove for hand rehabilitation,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, Tokyo, Japan, Nov. 3–7 2013, pp. 1512–1517. ([GMW Group](#))
- [4] Y. Wang, S. Zheng, Z. Song, J. Pang, and J. Li, “A coupling dynamic model for studying the physical interaction between a finger exoskeleton and a human finger,” *IEEE Access*, vol. 8, pp. 125 412–125 422, 2020. ([Academia](#))
- [5] “Cyberpunk Mechanical Armor [Red Edition],” *YouTube*, short-format video, uploaded 2024. [Online]. Available: <https://youtube.com/shorts/SEkaF21Pd1Q>. [Accessed: May 13, 2025].
- [6] BODY3D, “Articulated Hand,” *MyMiniFactory*, Oct. 3 2024. [Online]. Available: <https://www.myminifactory.com/object/3d-print-articulated-hand-415727>. [Accessed: May 13, 2025]. ([MyMiniFactory](#))
- [7] MIT Media Lab, “Biomechatronics — Applicant Information,” *media.mit.edu*, 2025. [Online]. Available: <https://www.media.mit.edu/groups/biomechatronics/applicant-information/>. [Accessed: May 13, 2025]. ([MIT Media Lab](#))
- [8] R. Shearman, “1957 — ‘Artificial Muscle’ — Joseph Laws McKibben (American),” *cyberneticzoo.com*, Apr. 8 2012. [Online]. Available: <https://cyberneticzoo.com/bionics/1957-artificial-muscle-joseph-laws-mckibben-american/>. [Accessed: May 13, 2025]. ([cyberneticzoo.com](#))
- [9] Voltera Inc., “Robotic Hand and Control Glove with Printed Stretchable Strain Sensors — NOVA Case Study,” technical white paper, 2022. [Online]. Available: https://uploads-ssl.webflow.com/.../2022_NOVA%20robot%20glove%20case%20study.pdf. [Accessed: May 13, 2025]. ([Webflow](#))
- [10] R. P. Weiss, “Multi-Material Experiments: TPU / PLA,” *Rainer’s World* (blog), Nov. 23 2019. [Online]. Available: <https://schlosshan.eu/blog/2019/11/23/multi-material->

[experiments-tpu-pla/](#). [Accessed: May 13, 2025].
([schlosshan.eu](#))

[11] L. Fried, “Using an FSR,” *Adafruit Learning System*, originally published Jul. 29 2012, last edited Mar. 8 2024. [Online]. Available: <https://learn.adafruit.com/force-sensitive-resistor-fsr/using-an-fsr>. [Accessed: May 13, 2025]. ([Adafruit Learning System](#))

[12] Texas Instruments, “INA180 — 26 V, 350 kHz current-sense amplifier,” product page, 2025. [Online]. Available: <https://www.ti.com/product/INA180>. [Accessed: May 13, 2025]. ([Texas Instruments](#))

[13] Texas Instruments, *INAx180 Low- and High-Side Voltage Output, Current-Sense Amplifiers* datasheet, rev. H, Jul. 2022. [Online]. Available: <https://www.ti.com/lit/ds/symlink/ina180.pdf>. [Accessed: May 13, 2025]. ([Texas Instruments](#))